

Hormuud University
Discrete Mathematics
Assignment 1

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1.Question 1

a) Users can log in only if the server is online and the database is accessible.

$$r \rightarrow (p \wedge q)$$

b) If the server is online but the database is not accessible, then users cannot log in.

$$(p \wedge \neg q) \rightarrow \neg r$$

c) Users can log in if and only if the server is online and the database is accessible.

$$r \leftrightarrow (p \wedge q)$$

d) Either the server is offline or users cannot log in.

$$\neg p \vee \neg r$$

e) If users can log in then the database must be accessible.

$$r \rightarrow q$$

Question 2 — Truth Tables

a)) $(p \vee q) \rightarrow r$

p	q	r	$p \vee q$	$p \vee q \rightarrow r$
T	T	T	T	T
T	T	F	T	F
T	F	T	T	T
T	F	F	T	F
F	T	T	T	T
F	T	F	T	F
F	F	T	F	T
F	F	F	F	T

b) $(p \rightarrow q) \leftrightarrow (\neg p \vee q)$

P	q	$p \rightarrow q$	$\neg p \vee q$	$(p \rightarrow q) \leftrightarrow (\neg p \vee q)$
T	T	T	T	T
T	F	F	F	T
F	T	T	T	T
F	F	T	T	T

c) $(p \vee q) \rightarrow (p \oplus q)$

p	q	$P \vee q$	$P \oplus q$	$(p \vee q) \rightarrow (p \oplus q)$
T	T	T	F	F
T	F	T	T	T
F	T	T	T	T
F	F	F	F	T

d) $(p \oplus q) \rightarrow (p \wedge q)$

p	q	$p \oplus q$	$p \wedge q$	$) (p \oplus q) \rightarrow (p \wedge q)$
T	T	F	T	T
T	F	T	F	F
F	T	T	F	F
F	F	F	F	T

e) $(p \vee q) \oplus (p \wedge q)$

P	q	$p \vee q$	$p \wedge q$	$p \vee q \oplus (p \wedge q)$
T	T	T	T	F
T	F	T	F	T
F	T	T	F	T
F	F	F	F	F

f) $(p \leftrightarrow q) \oplus (\neg p \leftrightarrow q)$

p	q	$p \leftrightarrow q$	$\neg p$	$(\neg p \leftrightarrow q)$	$(p \leftrightarrow q) \oplus (\neg p \leftrightarrow q)$
T	T	T	F	F	T
T	F	F	F	T	T
F	T	F	T	T	T
F	F	T	T	F	T

g) $(p \leftrightarrow q) \oplus (\neg p \leftrightarrow \neg r)$

P	q	R	$p \leftrightarrow q$	$(\neg p \leftrightarrow \neg r)$	$(p \leftrightarrow q) \oplus (\neg p \leftrightarrow \neg r)$
T	T	T	T	T	F
T	T	F	T	F	T
T	F	T	F	T	T
T	F	F	F	F	F
F	T	T	F	F	F
F	T	F	F	T	T
F	F	T	T	F	T
F	F	F	T	T	F

h) $(p \oplus q) \rightarrow (p \oplus \neg q)$

P	Q	$p \oplus q \rightarrow (p \oplus \neg q)$
T	T	T
T	F	F
F	T	F
F	F	T

Question 3 — Logical Equivalence

a) $p \leftrightarrow q$ and $(p \wedge q) \vee (\neg p \wedge \neg q)$

P	q	$P \leftrightarrow q$	$P \wedge q$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$	$p \wedge q \vee \neg p \wedge \neg q$	
T	T	T	T	F	F	F	T	
T	F	F	F	F	T	F	F	
F	T	F	F	T	F	F	F	
F	F	T	F	T	T	T	T	

Answer $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$ Are equivalent

b) $p \rightarrow q$ and $\neg q \rightarrow \neg p$

p	q	$p \rightarrow q$	$\neg p$	$\neg q$	a) $q \rightarrow \neg p$
T	T	T	F	F	T
T	F	F	F	T	F
F	T	T	T	F	T
F	F	T	T	T	T

$p \rightarrow q \equiv \neg q \rightarrow \neg p$

same expression ✓

c) $\neg p \leftrightarrow q$ and $p \leftrightarrow \neg q$

p	q	$\neg p$	$\neg q$	$\neg p \leftrightarrow q$	$p \leftrightarrow \neg q$
T	T	F	F	F	F
T	F	F	T	T	T
F	T	T	F	T	T
F	F	T	T	F	F

Are Equivalent

d) $\neg(p \oplus q)$ and $p \leftrightarrow q$

p	Q	$p \oplus q$	$\neg p \oplus q$	$p \leftrightarrow q$
T	T	F	T	T
T	F	T	F	F
F	T	T	F	F
F	F	F	T	T

$\neg(p \oplus q) \equiv p \leftrightarrow q$ Are Equivalent

e) $(p \rightarrow q) \wedge (p \rightarrow r)$ and $p \rightarrow (q \wedge r)$

p	q	r	$p \rightarrow q$	$p \rightarrow r$	$p \rightarrow q) \wedge (p \rightarrow r$	$p \wedge r$	$p \rightarrow q \wedge r$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	F
T	F	T	F	T	F	F	F
T	F	F	F	F	F	F	F
F	T	T	T	T	T	T	T
F	T	F	T	T	T	F	T
F	F	T	T	T	T	F	T
F	F	T	T	T	T	F	T

So $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$ are Equivalent

f) $(p \rightarrow r) \wedge (q \rightarrow r)$ and $(p \vee q) \rightarrow r$

P	Q	r	$P \rightarrow r$	$q \rightarrow r$	$p \vee q$	$(p \rightarrow r) \wedge (q \rightarrow r)$	$(p \vee q) \rightarrow r$
T	T	T	T	T	T	T	T
T	T	F	F	F	T	F	F
T	F	T	T	T	T	T	T
T	F	F	F	T	T	F	F
F	T	T	T	T	T	T	T
F	T	F	T	F	T	F	F
F	F	T	T	T	F	T	T
F	F	F	T	T	F	T	T

: they are Equivalent : $(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$

g) $(p \rightarrow q) \vee (p \rightarrow r)$ and $p \rightarrow (q \vee r)$

p	q	r	$p \rightarrow q$	$p \rightarrow r$	$(p \rightarrow q) \vee (p \rightarrow r)$	$q \vee r$	$p \rightarrow (q \vee r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	F	F	F	F
F	T	T	T	T	T	T	T
F	T	F	T	T	T	T	T
F	F	T	T	T	T	F	T
F	F	F	T	T	T	F	T

They are Equivalent : $(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$

h) $(p \rightarrow r) \vee (q \rightarrow r)$ and $(p \wedge q) \rightarrow r$

p	q	r	$p \rightarrow r$	$q \rightarrow r$	$(p \rightarrow r) \vee (q \rightarrow r)$	$p \wedge q$	$(p \wedge q) \rightarrow r$
T	T	T	T	T	T	T	T
T	T	F	F	F	F	T	F
T	F	T	T	T	T	F	T
T	F	F	F	T	T	F	T
F	T	T	T	T	T	F	T
F	T	F	T	F	T	F	T
f	F	T	T	T	T	F	T

F	F	F	T	T	T	F	T
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Are same $(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

i) $\neg p \rightarrow (q \rightarrow r)$ and $q \rightarrow (p \vee r)$

p	q	r	$q \rightarrow r$	$\neg p$	$\neg p \rightarrow (q \rightarrow r)$	$p \vee r$	$q \rightarrow (p \vee r)$
T	T	T	T	F	T	T	T
T	T	F	F	F	T	T	T
T	F	T	T	F	T	T	T
T	F	F	T	F	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	T	F	F	F
F	F	T	T	T	T	T	T
F	F	F	T	T	T	F	T

So they are Equavalent

j) $p \leftrightarrow q$ and $(p \rightarrow q) \wedge (q \rightarrow p)$

p	q	$p \leftrightarrow q$	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T	T
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

$p \leftrightarrow q$ and $(p \rightarrow q) \wedge (q \rightarrow p)$ are Equavalent

Question 4: Constructing Logical Statements

Let:

- p: You study discrete mathematics
- q: You complete the assignments
- r: You pass the course

a) If you study discrete mathematics and complete the assignments, then you will pass the course.

Logical expression : $(p \wedge q) \rightarrow r$

b) If you do not study discrete mathematics, then you will not pass the course.

Logical expression: $\neg p \rightarrow \neg r$

c) You pass the course if and only if you study discrete mathematics and complete the assignments.

Logical expression: $r \leftrightarrow (p \wedge q)$

d) You study discrete mathematics or you complete the assignments.

Logical expression: $p \vee q$